



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

DEHYDRATORS (E-2308A/B)

Docu. No. : 14780-8120-PS-E2308
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
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TITLE :

DEHYDRATORS

(E-2308A/B)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-E2308**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT		S.W.Park	J.S.Chung	- S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	- S.W.Lee
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REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW		APPROVAL
Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
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SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

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Case		Design	TEMA Type	AES	Orientation	Horizontal	Rev
Number of Units	2 (Note 2)		Shells per Unit	1	Shells in Series	1	
Surface per Unit	m ²	170.0	Surface per Shell	m ²	Shells in Parallel	1	
PERFORMANCE OF ONE UNIT							
		In	SHELLSIDE	Out	In	TUBESIDE	Out
Fluid Name		S4 Steam			Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O		
Total Flowrate (Note 3)		kg/h	3421 × 1.05		kg/h		61257 × 1.05
Vapour		kg/h	3421 × 1.05	0	kg/h		0
Liquid		kg/h	0	3421 × 1.05	kg/h		61257 × 1.05
Fluid Condensed		kg/h	3421 × 1.05		kg/h		0
Fluid Evaporated		kg/h	0		kg/h		0
LIQUID PROPERTIES							
		AS WATER		SEE PAGE 7			
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES							
		AS WATER					
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	146.3	144.8	75.0	140.0		
Operating Pressure	kg/cm ² g	3.185		6.221			
Number of Passes per Shell		1		6			
Average Velocity	m/s	3.02		1.35			
Pressure Drop (Allow / Calc)	kg/cm ²	Nil	0.024	0.703	0.589		
Fouling Factor	h.m ² .°C/kcal	0.0001		0.0010			
Duty (Note 3)	kcal/h × 10 ⁶	1.744 × 1.05		MTD	°C	23.3	
U-value Service/Clean	kcal/h.m ² .°C	474	1255	Over Surface	%	2.6	
DESIGN CONDITIONS							
		SHELLSIDE		TUBESIDE			
Design pressure max/min	kg/cm ² g	16.5		21.0			
Full Vacuum		Required	at 195 °C	Required	at 140 °C		
Test Pressure	kg/cm ² g	To Code		To Code			
Design Temperature max/min	°C	195	15	250	15		
Corrosion Allowance	mm	3		1.5			
Insulation Details		Heat Conservation		Heat Conservation			
Hydrogen Service		No		No			
Sour Service		No		No			
Lethal Service		No		No			
Cyclic Service		No		No			
Cleaning Method		Mechanical		Mechanical			
Stress Relief for Process Reasons		No		No			
Heat Release Curve		No		Yes			



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CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm	760	Channel I.D. (If different)	mm	
TUBES	Number of Tubes		486	Tube Length	mm	6096
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded	Tube-Baffle Diam. Clearance	mm	0.8
			(Note 5)			
BAFFLES	Number of Baffles		17	Baffle Type		Single Seg.
	Baffle Cut	%	26.61	Baffle Spacing	mm	300
	Baffle Cut Orientation		Vertical	Double Seg. Baffle Overlap	mm	0
	Baffle Thick	mm	6.4	Baffle-Shell Diam. Clearance	mm	4.8
BUNDLE	Impingement Protection		Circular	Tube Supports (Note 6)		At Floating Head
	Pairs of Sealing Strips		2	Sealing Strip Details (Note 8)		First to Last Baffle
	Passlane Sealing		Yes	Passlane Sealing Details (Note 8)		First to Last Baffle
	U-Bend Diam.	mm	N.A.	U-Bend Orientation		N.A.
	U-Bend Area Included		N.A.	U-Bend Support Details		N.A.
	Longitudinal baffle		N.A.	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE

Service	No.	N.B.		Rating	Facing
Inlet S1	1	8	Inch	300	RFWN
Outlet S2	1	3	Inch	300	RFWN
Non-condensibles Vent SV	1	2	Inch	300	RFLWN
Drain SD	1	2	Inch	300	RFLWN

TUBESIDE

Service	No.	N.B.		Rating	Facing
Inlet T1	1	6	Inch	300	RFWN
Outlet T2	1	6	Inch	300	RFWN
Vent TV	1	3/4	Inch	300	RFLWN
Drain TD	1	3/4	Inch	300	RFLWN

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	Killed Carbon Steel (UOP Std 3-11)
Floating Head Cover	304L Stainless Steel (UOP Std 3-15)
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	304L Stainless Steel
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: Killed Carbon Steel
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell (Tonnes)	Exchanger Weight, Dry	7.0	Exchanger Weight, Wet	9.9
	Bundle Weight, Dry	2.7		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-11-7, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

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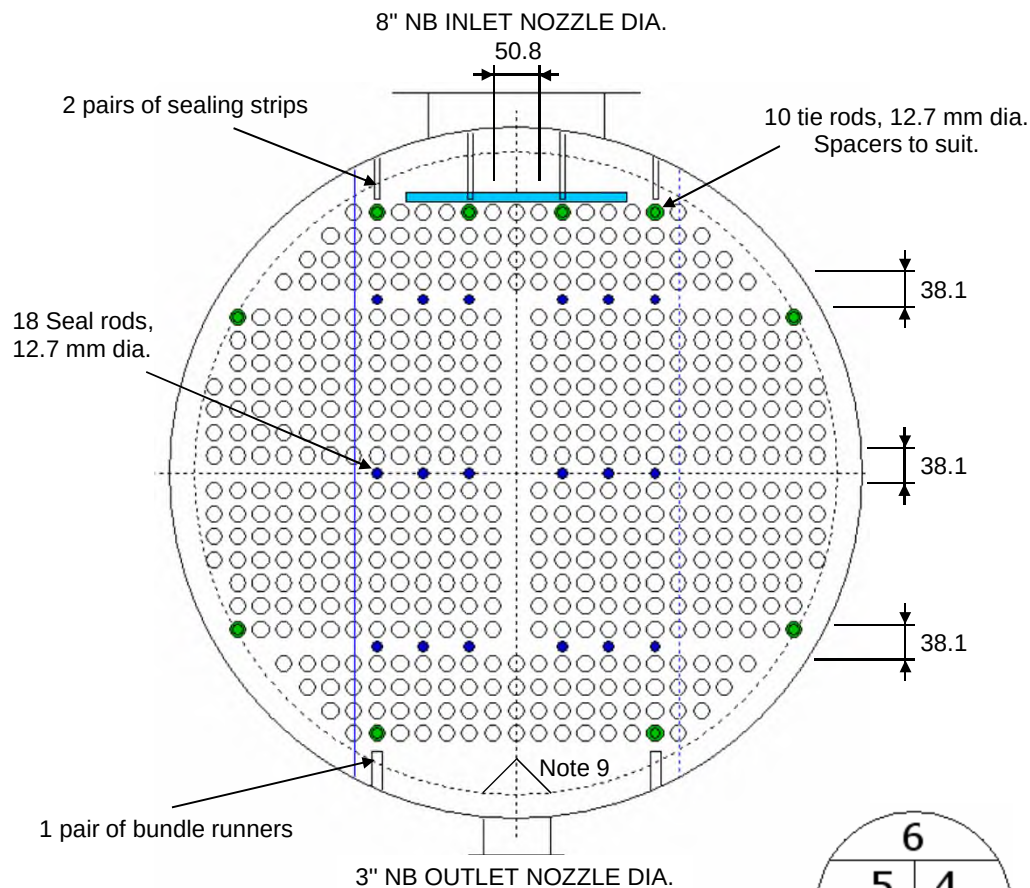
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Notes:

1. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
2. Two identical heat exchangers in parallel are required. One is for normal operation and one is for a spare.
3. Flow and duty include 5% design margin.
4. Alternate design steam conditions at inlet are 4.5 kg/cm²g at 165°C. Saturation temperature is 155°C.
5. Strength-weld tubes to tubesheet per requirements on page 8.
6. Provide a floating head support plate.
7. Bundle runners shall not restrict the shellside exit flow path.
8. Sealing strips, bundle runners, and Seal rods shall not restrict the bundle cleaning lanes.
9. All baffles and support plates extending to the bottom of the shell shall be provided with V-notch at the lowest point.
10. Minimise the volume of the process material in the tubeside. VENDOR to confirm.
11. Test ring shall be provided by VENDOR.
12. Deleted.

Tube Layout



All dimension in mm
Sketch not to scale

Tubepass Arrangement

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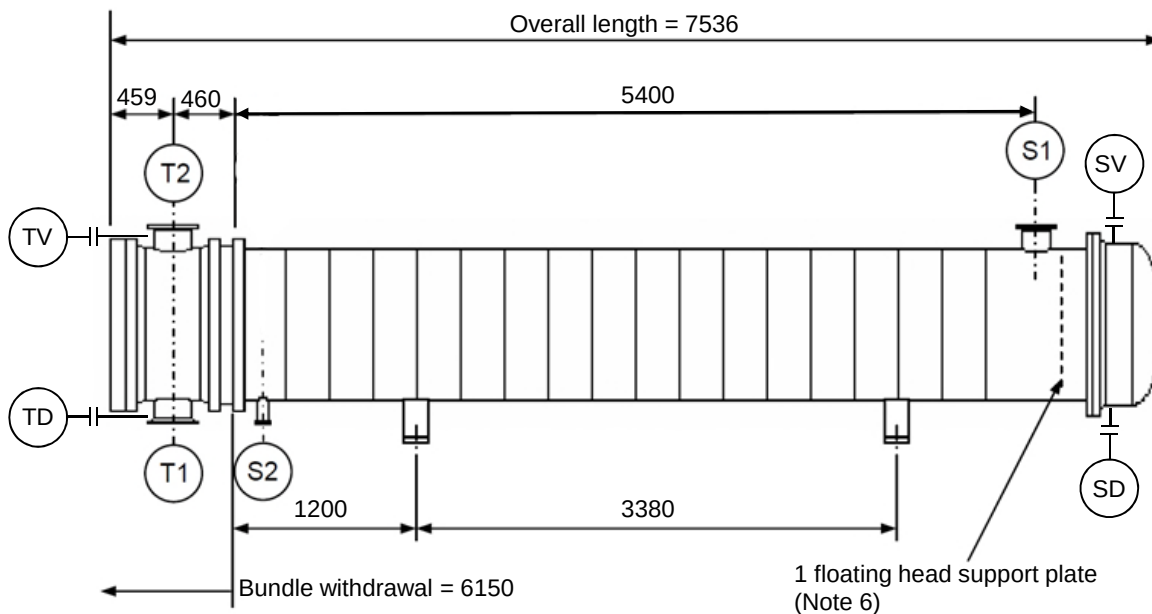
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Setting Plan

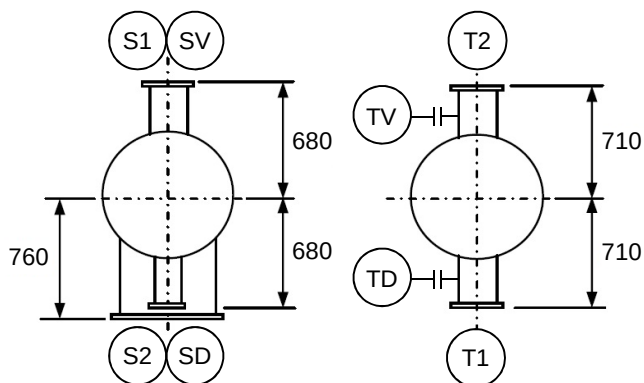


Shell ID : 760 mm

Tube length : 6096 mm

Shellside Nozzles

Tubeside Nozzles



PROCESS NOZZLE SUMMARY

REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	S3 Steam Inlet	8"	300#RFLWN		
S2	S3 Steam Outlet	3"	300#RFLWN		
T1	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Inlet	6"	300#RFLWN		
T2	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Outlet	6"	300#RFLWN		
SV	Non-condensibles Vent	2"	300#RFLWN		
SD	Shellside Drain	2"	300#RFLWN		CW/blind
TV	Tubeside Vent	3/4"	300#RFLWN		CW/blind
TD	Tubeside Drain	3/4"	300#RFLWN		CW/blind

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ColdSide Property Curve

Pressure

6.221

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
75.0	0.00	0.0000	-	-	-	-
82.8	3.16	0.0000	-	-	-	-
90.5	6.32	0.0000	-	-	-	-
98.0	9.48	0.0000	-	-	-	-
105.4	12.64	0.0000	-	-	-	-
112.6	15.80	0.0000	-	-	-	-
119.7	18.97	0.0000	-	-	-	-
126.6	22.13	0.0000	-	-	-	-
133.4	25.29	0.0000	-	-	-	-
140.0	28.45	0.0000	-	-	-	-

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
75.0	900.3	0.600	0.120	0.51	27.5	59.7	336
82.8	891.7	0.536	0.119	0.52	26.6	59.7	336
90.5	883.2	0.483	0.117	0.53	25.7	59.8	336
98.0	874.7	0.440	0.116	0.54	24.8	59.8	336
105.4	866.3	0.405	0.114	0.55	24.0	59.8	336
112.6	857.9	0.375	0.113	0.56	23.1	59.9	336
119.7	849.6	0.349	0.111	0.57	22.3	59.9	335
126.6	841.3	0.328	0.110	0.58	21.5	59.9	335
133.4	833.1	0.309	0.109	0.60	20.8	60.0	335
140.0	824.9	0.293	0.107	0.61	20.0	60.0	335

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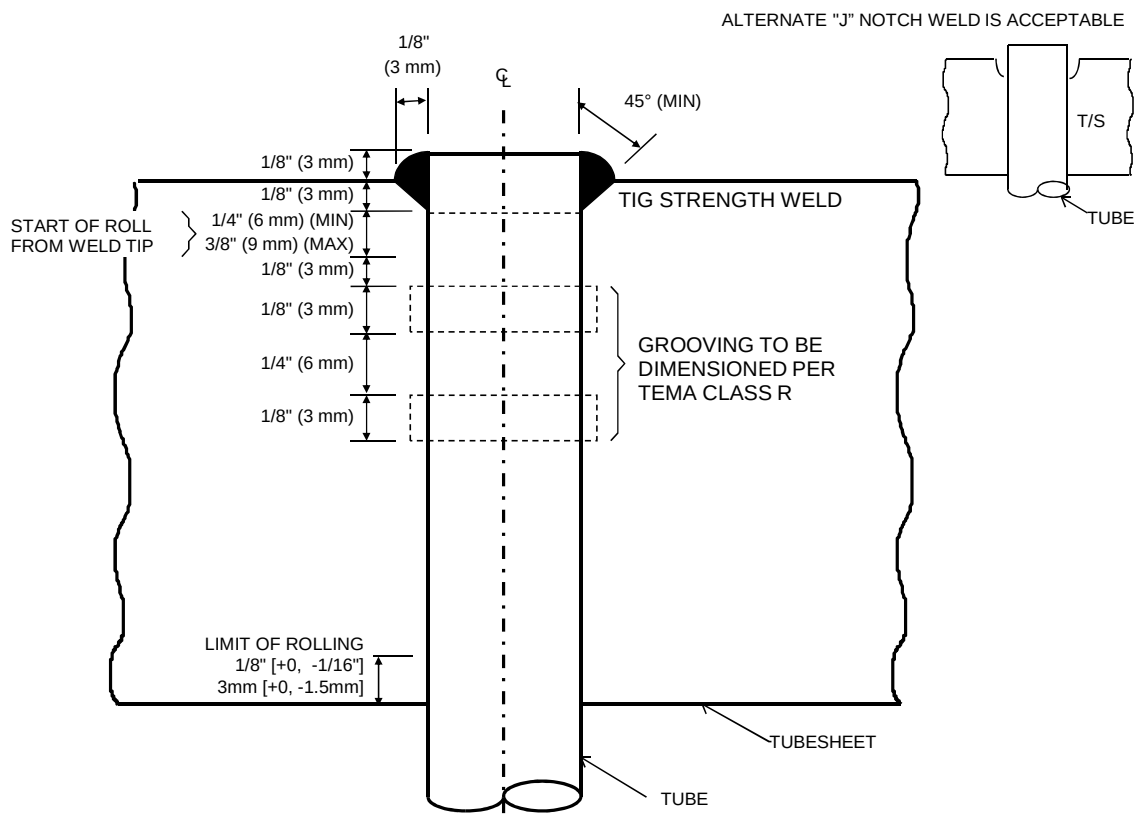
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Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

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